

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281823

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

TSAI; Chi-Ming et al.

FAST-RADIATING CLAW ASSEMBLY OF A CLAW MACHINE

Abstract

A fast-radiating claw assembly includes a cylinder, a piston movable in the cylinder, claws pivotally connected to the piston, levers for pivotally connecting the claws to the cylinder, a solenoid unit operable for moving the piston, and a metal net for covering at least a portion of a periphery of the solenoid unit. The net includes a side in contact with the cylinder and another side in contact with the solenoid unit.

Inventors: TSAI; Chi-Ming (Hemei Town, TW), YANG; I-Chiang (Hemei Town, TW)

Applicant: FEILOLI ELECTRONIC CO., LTD. (Hemei Town, TW)

Family ID: 1000007728458

Appl. No.: 18/601313

Filed: March 11, 2024

Publication Classification

Int. Cl.: A63F9/30 (20060101)

U.S. Cl.:

CPC A63F9/30 (20130101);

Background/Summary

BACKGROUND OF INVENTION

1. Field of Invention

[0001] The present invention relates to a claw assembly of a claw machine and, more particularly, to a fast-radiating claw assembly of a claw machine.

2. Related Prior Art

[0002] Referring to FIG. 1, a conventional claw machine includes a conventional claw assembly equipped with a conventional solenoid unit. The conventional solenoid unit includes a cylinder or shell **10** that contains tubular core **11** that extends throughout a solenoid **12**. The solenoid **12** generates magnetism when electricity goes through it. A claw assembly **13** is pivotally connected to the cylinder **10** via a linkage (not numbered) and a piston **14**. The claw assembly **13** includes three claws for example. The piston **14** is movable in the tubular core **11**. When electricity goes through the solenoid **12**, it generates magnetism to lift the piston **14** to close the claw assembly **13**. When electricity does not go through the solenoid **12**, it stops magnetism to allow the piston **14** to fall to open the claw assembly **13**.

[0003] As the claw assembly is in operation, the solenoid **12** generates heat as a byproduct. The temperature of the solenoid **12** inevitably rises and the electricity drops and so does the magnetism.

[0004] Various solutions have been devised for the above-mentioned problem. For example, vents can be made in the cylinder, a fan can be used to expel heat via vents made in the cylinder, or a radiating layer can be used around the solenoid. These solutions are however not satisfactory.

[0005] The present invention is therefore intended to obviate or at least alleviate the problems encountered in the prior art.

SUMMARY OF INVENTION

[0006] It is the primary objective of the present invention to provide a fast-radiating claw assembly.

[0007] To achieve the foregoing objectives, the fast-radiating claw assembly includes a cylinder, a piston movable in the cylinder, claws pivotally connected to the piston, levers for pivotally connecting the claws to the cylinder, a solenoid unit operable for moving the piston, and a metal net for covering at least a portion of a periphery of the solenoid unit. The net includes a side in contact with the cylinder and another side in contact with the solenoid unit.

[0008] Other objectives, advantages and features of the present invention will be apparent from the following description referring to the attached drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] The present invention will be described via detailed illustration of the preferred embodiment referring to the drawings wherein:

[0010] FIG. 1 is an exploded view of a conventional claw assembly;

[0011] FIG. 2 is an exploded view of a fast-radiating claw assembly according to the preferred embodiment of the present invention;

[0012] FIG. 3 is a perspective view of a solenoid unit of the claw assembly shown in FIG. 2;

[0013] FIG. 4 is an enlarged partial view of a net of the solenoid unit shown in FIG. 3;

[0014] FIG. 5 is a perspective view of the claw assembly shown in FIG. 2; and

[0015] FIG. 6 is a cross-sectional view of the claw assembly taken along a line VI-VI shown in FIG. 5.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

[0016] Referring to FIGS. 2 through 6, a claw machine can include a fast-radiating claw assembly that includes a cylinder **20**, a solenoid unit **22**, a washer **23**, a lid **24**, claws **25**, levers **26**, a piston **27** and a metal net **30** according to the preferred embodiment of the present invention. The cylinder **20** is made with vents **21** to facilitate heat to exit the cylinder **20**. The claws **25** are pivotally connected to the piston **27**, which is movable in the cylinder **20**. Each of levers **26** includes an end pivotally connected to a corresponding one of the claws **25** and another end pivotally connected to the cylinder **20**.

[0017] The solenoid unit **22** includes at least one solenoid. The solenoid unit **22** is located in the

cylinder **20** so that the solenoid unit **22** extends around the piston **27**. The washer **23** is located between the lid **24** and an end of the solenoid unit **22** when the lid **24** is connected to the cylinder **20**.

[0018] When electricity travels through the solenoid unit **22**, it generates magnetism to lift the piston **27** to close the claws **25**. When electricity does not travel through the solenoid unit **22**, it stops magnetism to allow the piston **27** to fall to open the claws **25**.

[0019] The metal net **30** extends around the solenoid unit **22**. The metal net **30** covers at least one second (50%) of the periphery of the solenoid unit **22**. Preferably, the metal net **30** covers all the periphery of the solenoid unit **22**. The metal net **30** metal wires **31** extend in a crisscross manner, thereby providing meshes **32**. The total area of the meshes **32** in the metal net **30** is larger than the total area of the metal wires **31** in the metal net **30**. The metal net **30** includes an internal side in contact with the solenoid unit **22** and an external side in contact with the cylinder **20** so that there is thermally no gap between the solenoid unit **22** and the cylinder **20**. In operation, heat is transferred to the cylinder **20** from the solenoid unit **22**. The heat is then radiated from the cylinder **20**.

[0020] The metal net **30** is sandwiched between the solenoid unit **22** and the cylinder **20** because conduction of heat to the cylinder **20** from the solenoid unit **22** via the metal net **30** is faster than convection of heat to the cylinder **20** from the solenoid unit **22**. Hence, the heat leaves the solenoid unit **22** fast so that the temperature of the solenoid unit **22** does not rise considerably so that the electricity that goes through the solenoid unit **22** does not drop considerably and nor does the magnetism. Therefore, the piston **27** is always adequately lifted and hence the claws **25** are always adequately closed.

[0021] Two metal nets **30** or more can be used around the solenoid unit **22** if so desired.

[0022] The present invention has been described via the illustration of the preferred embodiment. Those skilled in the art can derive variations from the preferred embodiment without departing from the scope of the present invention. Therefore, the preferred embodiment shall not limit the scope of the present invention defined in the claims.

Claims

1. A fast-radiating claw assembly comprising a cylinder (**20**), a piston (**27**) movable in the cylinder (**20**), claws (**25**) pivotally connected to the piston (**27**), levers (**26**) for pivotally connecting the claws (**25**) to the cylinder (**20**), a solenoid unit (**22**) operable for moving the piston (**27**), and a metal net (**30**) for covering at least a portion of a periphery of the solenoid unit (**22**), wherein the net (**30**) comprises a side in contact with the cylinder (**20**) and another side in contact with the solenoid unit (**22**).
 2. The fast-radiating claw assembly according to claim 1, wherein the metal net (**30**) comprises metal wires (**31**) extending in a crisscross manner, thereby providing meshes (**32**).
 3. The fast-radiating claw assembly according to claim 1, wherein a total area of the meshes (**32**) in the metal net (**30**) is larger than a total area of the metal wires (**31**) in the metal net (**30**).
 4. The fast-radiating claw assembly according to claim 1, wherein the metal net (**30**) covers at least 50% of the periphery of the solenoid unit (**22**).
 5. The fast-radiating claw assembly according to claim 4, wherein the metal net (**30**) covers all the periphery of the solenoid unit (**22**).
 6. The fast-radiating claw assembly according to claim 1, wherein the cylinder (**20**) comprises vents (**21**) for facilitating heat to exit the cylinder (**20**).
-